

18 Sep. 2026 存算一体在AI中的应用以及未来发展 赖梁祯

Compute-In-Memory in AI Application and Its Future Development

Computational Power Demands of AI

5B Speech and multimodal input/output
30B Summarizing
112B Reasoning
800B General-purpose embodied intelligence
1T Producing.....

Evolution Trend of Large Models

Demands varies among different applications
Prefill speed
Prefill tokens
Context length

CIM

Analog: accuracy & noise -> requires algorithms to refine

Digital: more flexible

Multiply based on Ohm's law: $I = G * V$

RC delay & Accuracy: If the scale gets bigger, the computing efficiency ascends but the delay gets longer and the accuracy declines. (Wirelength)

Future Development

NAND flash, MRAM, **SRAM(many chips have been using)**, Nor flash, Feram, RRAM.....
(different medias)

Speed -> RC AD/DA Area -> architecture design optimizing

Accuracy: the gate capacity of electron is limited -> Improving manufacturability

Density: 28nm/40nm -> New media

Write in/Erase..... -> New Media

3D/2.5D advanced packaging:

Pros:

Distinctively integrated: inclusive for different logic, memory and CIM

Improve capacity, computational power

Cons:

Energy supply, power, heat. Media can be sensitive to temperature and noise

Cooperating with algorithms

Balancing delays between CIM and non-CIM

Q&A:

1. By now only simpler things like matrix multiplication can be achieved by CIM (directly), other complex things are able to be tackled through software.
2. Computing demands massive and intricate manipulating of cells in memory.
3. CIM excels in decoding, but this doesn't mean it is weak at the stage of prefilling.
4. Multi-core manipulating in CIM for AI (I mean the researching process) is similar to what we have done in GPU.

